

Construction of Driving Behavior Scoring Model Based on OBD Terminal Data Analysis

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Abstract—In order to accurately identify abnormal driving behaviors, this paper proposes different abnormal driving behavior recognition algorithms. The algorithm obtains driver's driving data through the OBD terminal, combines the x-axis and y-axis acceleration changes and behavior duration of the vehicle's three-axis acceleration sensor to identify abnormal driving behaviors, establishes a hierarchical driving behavior indicator system and judgment matrix. On this basis, a driving behavior scoring model is established. The model combines the driving data of the driver, takes the proportion of abnormal driving behavior as the evaluation index, and uses the entropy weight method and the analytic hierarchy process to obtain the index weight. The test results of an example show that the model can analyze and evaluate the driving behavior of the driver well, and give a score of driver's behavior, to encourage the driver to develop good driving habits and effectively prevent the occurrence of traffic accidents.

Keywords—On board diagnostics; driving behavior; entropy method; analytic hierarchy process

I. INTRODUCTION

In recent years, with the rapid economic development, our country's automobile industry has developed vigorously. At the same time, the number of national vehicles has continued to grow, which has brought a series of road traffic accidents and casualties. Researches shows that more than 90% of large-scale traffic accidents in our country are caused by driver's personal reasons. Therefore, researching on driver's driving behaviors is very important.

In the process of researching driver's driving behaviors, in terms of driving behavior analysis, because novice drivers lack relevant driving system knowledge, dangerous driving behaviors are prone to cause considerable harm to traffic safety, personal life and property safety. Huiping Zhou pointed out the limitations of driver perception and system response, and has important guiding significance for accidents caused by system failures[1]. Bian Yang proposed a related measure for pedestrian safety and proposed behavior model theory[2]. Mohammad Shahverdy pointed out that although the monitoring of driver behavior has been widely used, its accuracy and safety still have hidden dangers. He proposed a more effective monitoring of the driver's behavior by multi-layer indicators including acceleration, deceleration, gravity and throttle[3].

Drivers will have many abnormal driving behaviors during traffic driving, such as rapid acceleration, rapid deceleration, and sharp turns. These are all aggressive driving behaviors, which

are one of the main reasons leading to traffic accidents. Stephen J. Maybank has studied a detection model for abnormal driving behavior, which makes the detection of abnormal driving behaviors more accurate and complete[4]. Wuqiang Meng detects abnormal driving behaviors by analyzing more standardized driving behaviors, and establishes a driver model to monitor, control speed and quantitatively evaluates abnormal driving behaviors[5]. At present, the Internet of Vehicles industry is developing rapidly. Most cars are equipped with OBD equipment[6], which can detect vehicle speed, acceleration, engine speed and acceleration data of the vehicle's three-axis accelerometer, analyze and process the collected data, and find out where the driver is some problems in the driving process can effectively prevent traffic accidents. Based on the related technologies of the Internet of Vehicles, this paper combined with the driving data obtained by OBD[7], and analyzed the relevant abnormal driving behavior of the driver in detail. The driver's driving behavior scoring method is established to score the driver's driving behavior.

To sum up, the driver is the most important component in the urban traffic system. Grasping the characteristics of the driver's driving behavior is the key to solving existing traffic problems. The research on driving behavior has an impact on the reduction and life of traffic accidents. The loss of property is of great significance.

II. ABNORMAL DRIVING BEHAVIOR RECOGNITION ALGORITHM

The various driving operations of the driver during the driving process can obtain relevant data through the OBD equipment loaded on the vehicle, and realize the identification of various abnormal driving behaviors through the processing of these data. CAN bus data, three-axis acceleration data and GPS satellite data are collected by the OBD terminal for in-depth mining and analysis of driver's driving behavior.

A. Speeding behavior recognition algorithm

In order to facilitate the statistics of the number of driver speeding behaviors, this paper constructs a speeding behavior recognition algorithm. On the basis of obtaining the speed of the OBD data dashboard, the maximum speed limit of each road section is set as the overspeed threshold V_s . According to the speed limit of the vehicle at a general high speed, the overspeed threshold is taken here as 80Km/h, and the overspeed limit is introduced. T_s is 2s.

The algorithm flow of speeding behavior recognition is as follows:

Step1, Obtain driver driving data.

Step2, Determine whether the speed V_t is greater than the overspeed threshold V_s , if it is greater, record the data collection time, continue to determine the speed V_t , and record the duration; if it is not greater, no overspeed behavior has occurred, and go to step 3.

Step3, Determine whether the overspeed duration T_c is greater than or equal to T_s , and if so, the number of overspeed behaviors is accumulated by 1.

Step4, Judge whether it is the last piece of data in the itinerary, if otherwise, return to step 2, if yes, output the number of overspeed, and the algorithm ends.

B. Rapidly accelerated behavior recognition algorithm

Since the judgment of the rapid acceleration behavior only involves the output value of the Y-axis of the three-axis accelerometer, only the a_y value in the three-axis acceleration data is processed. Introduce the threshold S_a for rapid acceleration behavior, and set the duration of acceleration behavior T_a to 2s, and pre-fetch the value of S_a according to industry experience.

$$S_a = 3m/s^2 \quad (1)$$

The flow of the rapid acceleration behavior recognition algorithm is as follows:

Step1, Obtain driver driving data.

Step2, Determine whether a_y is greater than the threshold S_a , if it is greater, record the collection time of this piece of data, continue to judge a_y and record the duration; otherwise, no sudden acceleration behavior occurs, and go to step 3.

Step3, Determine whether the rapid acceleration duration T_c is greater than or equal to T_a , if so, the number of rapid acceleration behaviors is accumulated by 1.

Step4, Judge whether it is the last piece of data in the itinerary, if otherwise, return to step 2, otherwise output the number of rapid acceleration and the algorithm ends.

C. Recognition algorithm for sudden deceleration and sudden braking

Since the judgment of sudden deceleration only involves the acceleration value in the Y direction, only the acceleration value in the Y direction is processed. The acceleration threshold of sudden deceleration to $-3m/s^2$, and set the acceleration threshold of sudden braking to $-4m/s^2$. And when the vehicle stops moving within a certain period of time, it is a sudden braking behavior.

TABLE I. BEHAVIOR JUDGMENT CONDITIONS

behavior	Acceleration	duration
Slow down	$a_y \leq 3m/s^2$	$T > 1s$
Brake hard	$a_y \leq -4m/s^2$	none

When the Y-axis acceleration is less than $-4m/s^2$, judge whether the driving speed V of the vehicle is less than $0.5m/s$, if so, the driving behavior is judged to be a sudden braking behavior.

D. Sharp turning behavior recognition algorithm

The sharp turn behavior is related to the X-axis acceleration of the three-axis accelerometer, so the sharp turn threshold and duration T are introduced. According to industry experience, T is 2s. Under normal circumstances, road bumps and other behaviors such as lane changes are excluded, and the sharp turn threshold is introduced.

$$|a_x| \geq 2m/s^2 \quad (2)$$

When this condition is met, it is judged whether the duration is greater than T , if so, the driving behavior is judged as a sharp turn behavior, the number of sharp turns is increased by 1, and the final output is output.

III. SCORING MODEL ESTABLISHMENT

A. The calculation principle of scoring index weight

1) The Entropy weight method

The theoretical basis of the entropy method is to determine the objective weight according to the variability of the index. Generally speaking, the lower the information entropy of an indicator, the higher the degree of variation of the indicator value, the more information it can contribute, and the greater the role it plays in the comprehensive evaluation, the greater the corresponding weight. On the contrary, the higher the information entropy of a certain index, the lower the degree of variation of the index value, the less information it can provide, the smaller the role in the comprehensive evaluation, the smaller the weight.

2) The Analytic Hierarchy Process

The analytic hierarchy process is a subjective weighting method, which mainly depends on a weighting method calculated by the decision maker to compare the importance of each index and assign values; the analytic hierarchy process decomposes elements into a hierarchical structure such as goals, criteria, and indicators, Compare the importance of indicators at the same level through the judgment matrix $B = (b_{ij})_{m \times n}$, where b_{ij} is the ratio of the indicators X_i to X_j , and finally calculate the weight of each indicator according to the judgment matrix.

3) Comprehensive Index Analysis

The entropy method starts from real data and has a high degree of objectivity. The analytic hierarchy process is a subjective weighting method, in which the judgment matrix is determined one by one based on human experience. The more difficult problems are broken down into simple factors, and then a hierarchical structure is constructed, using a pairwise comparison method. Determine the relative importance of decisions. Combining the advantages of the two, not only has the support of objective data, but also can simplify complex issues to calculate the index weight.

B. Construction of scoring index system

In order to highlight the scoring effect, the monthly behavior statistics are used as the analysis basis for driving behavior scoring, and the number of vehicle speeding, sharp shifts, sharp turns and abnormal speeds are counted based on the real-time data collected by the vehicle terminal.

According to the principles of systematic, independence, objectivity and maneuverability, 1 first-level indicator, 2 second-level indicators, and 6 third-level indicators are selected as the main evaluation factors for comprehensive scoring of driving behavior. The first level indicator is the comprehensive score of the driver's driving behavior. The second-level indicators include driving speed and four-speed driving behavior. Among them, the third-level indicators of driving speed include the average vehicle speed and the number of driving speeding. The four-speed driving behavior indicators are the rapid acceleration, rapid deceleration, rapid braking, and The number of sharp turns is evaluated as shown in the Table II.

TABLE II. HIERARCHICAL TABLE OF DRIVING BEHAVIOR INDICATORS

Target layer	standard layer α	Index layer β
Comprehensive Driver Behavior Score θ	Driving speed θ_1	Speeding times θ_{11}
		Average speed θ_{12}
	Four anxious behaviors θ_2	Rapid acceleration θ_{21}
		Rapid decelerations θ_{22}
		Sudden stops times θ_{23}
		Sharp turns times θ_{24}

C. Determining the weight of scoring index

First, calculate the standard layer weight and index layer weight of the analytic hierarchy process, then calculate the index weight of the entropy method, and standardize the original data before calculation. The comprehensive weights of each index of the criterion layer and factor layer are obtained respectively. The weight results calculated by the three methods of analytic hierarchy process, entropy weight method and comprehensive index weight analysis are shown in the Table III.

TABLE III. COMPARISON OF WEIGHT RESULTS OF DIFFERENT METHODS

α	β	AHP Weights		EWM Weights		Combination	
θ_1	θ_{11}	0.2666	0.3333	0.0912	0.1082	0.1055	0.1057
	θ_{12}	0.0667		0.0170		0.0002	
θ_2	θ_{21}	0.0456	0.6667	0.1901	0.8918	0.0134	0.8943
	θ_{22}	0.2041		0.2509		0.4683	
	θ_{23}	0.3035		0.0673		0.0745	
	θ_{24}	0.1135		0.3834		0.3381	

According to the weight calculation results in Table 3.2, the weight of the driving speed calculated by the analytic hierarchy process is 0.3333, and the weight of the four emergency behaviors is 0.6667; the weight of the driving speed calculated by the entropy weight method is 0.1082, and the weight of the four emergency behavior is 0.1082. The weight is 0.8918; the

weight of the driving speed calculated by combining the analytic hierarchy process and the entropy method is 0.1057, and the weight of the four emergency behaviors is 0.8943. Since the second-level indicators of driving speed include the number of speeding and average speed, and the second-level indicators of four-speed behavior include rapid acceleration, rapid deceleration, rapid braking and sharp turns, to a significant degree, the four-speed behavior is far greater than driving speed, and Among the four emergency behaviors, the importance of the number of sudden decelerations is the highest, followed by sharp turns, sudden braking and sudden acceleration.

In summary, the index weight calculated by the comprehensive index evaluation method combining the analytic hierarchy process and the entropy weight method is used as the evaluation index of the driving behavior scoring model, and the score adopts a 100-point system.

D. Comprehensive driving behavior score

To quantitatively score the driving behavior safety level, reasonable parameter values need to be set, and a large amount of driving behavior data statistics and analysis are required to determine the alternatives of each indicator in the driving behavior score index. Therefore, when constructing the driving behavior scoring model, first perform statistical analysis on the sample data of each factor, make different analysis for different models, occupations, etc., and combine the opinions of experts to obtain the alternatives and corresponding driving behavior scoring models. Score, establish a driving behavior scoring model as shown in Table IV.

TABLE IV. DRIVING BEHAVIOR SCORING MODEL TABLE

α	β	Scoring criteria
θ_1	θ_{11}	The ratio of the number of speeding to the total amount of data is less than 0.1 to get 90 points, and more than 70 points.
	θ_{12}	Average speed is less than 25km/h get 100 points, 25-45 km/h gets 90 points, 45-65 km/h gets 75 points, 65 km/h or more gets 55 points.
θ_2	θ_{21}	The ratio of the number of times to the total amount of data is 90 points below 0.1, 70 points between 0.1-0.15, 60 points between 0.15-0.2, and 40 points above 0.2
	θ_{22}	
	θ_{23}	
	θ_{24}	The ratio of the number of sudden stops to the total amount of data is less than 0.01 to get 85 points, between 0.01 and 0.02 to get 70 points, and more than 0.02 to get 65 points.

IV. MODEL CASE ANALYSIS

This paper selects part of the driver's driving statistics from the OBD service platform as the basis for analysis, and counts the number of speeding behaviors, rapid accelerations, rapid decelerations, rapid braking and sharp turns. According to the data obtained by OBD, the total amount of data of the driver 668304050264877 is 8368. The statistics of the number of actions are shown in Table V.

TABLE V. DRIVER'S DRIVING BEHAVIOR STATISTICS TABLE

Driver	Speeding times	Average speed	Rapid acceleration
8304050264877	197	29	919
Rapid decelerations	Sudden stops times	Sharp turns times	Total data
863	182	1095	8368

Since the total amount of data obtained for each driver is different, the number of rapid acceleration, rapid deceleration, and sharp turns also corresponds to different indicators. Therefore, we need to calculate the proportion of each driver's corresponding total data. According to the proportion to measure the weight of the driver under different indicators. Different driving behaviors under different indicators have different scores. At this time, the evaluation function of the driver's score is following.

$$\text{score} = w_1\theta_{11} + w_2\theta_{12} + w_3\theta_{21} + \dots + w_6\theta_{24} \quad (3)$$

The following table shows the comprehensive score of the driver passing the evaluation function.

TABLE VI. THE PROPORTIONS AND COMPREHENSIVE SCORES OF SOME DRIVERS' DRIVING BEHAVIORS

Driver	Speeding	Rapid acceleration	Rapid decelerations
8304050264877	0.035	0.109	0.103
Sudden stops	Sharp turns	Comprehensive score(points)	
0.0217	0.131	70	

Table VI shows the driver's driving behavior score under the driving behavior evaluation index established in this paper. Based on the comprehensive weights, it can be seen that the weights of rapid deceleration and sharp turning behaviors are relatively high. Therefore, the driver's driving behavior includes the rapid deceleration behavior and the driver with more sharp turns. Lower. In summary, the driving behavior score can have a quantitative indicator of the driver's driving behavior over a period of time, which can effectively reduce the number of

abnormal driving behaviors of the driver during the driving process.

V. SUMMARY

The rapid development of the Internet of Vehicles has brought effective guarantees for traffic safety, and at the same time provided practical and reliable technical solutions for the analysis of drivers' driving behaviors. This makes the collection, processing and analysis of driving behavior safety data convenient and quick. This paper mainly studies the driving behavior scoring model based on entropy weight method and analytic hierarchy process, analyzes driving behavior data by establishing a driving behavior recognition algorithm, and uses a comprehensive index weight determination algorithm to determine the weight of each index. This paper selects part of the real driving data of drivers, uses a driving scoring model to score their driving behavior, and verifies the reliability of the constructed driving scoring model with case analysis.

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REFERENCES

- [1] Yuan, Y., Lu, Y., Wang, Q. (2020) Adaptive forward vehicle collision warning based on driving behavior. *Neurocomputing*, 408:64-71.
- [2] Yang B., Liang K., Zhao X., Yang, L., Qu, W. (2020) Psychological influences on drivers' yielding behavior at the crosswalk of intersections. *Cognition, Technology & Work*, 22(3): 501-516.
- [3] Shahverdy, M., Fathy, M., Berangi, R., Sabokrou, M. (2020) Driver behavior detection and classification using deep convolutional neural networks. *Expert Systems with Applications*. 149: 113240.
- [4] Hu, J., Zhang, X., Maybank, S. (2020) Abnormal Driving Detection With Normalized Driving Behavior Data: A Deep Learning Approach. *IEEE Transactions on Vehicular Technology*. 69(7): 6943-6951.
- [5] Hu, J., Li, X., He, X., Meng, W. (2017) Abnormal Driving Detection Based on Normalized Driving Behavior. *IEEE Transactions on Vehicular Technology*. 66(8): 6645-6652.
- [6] Pereira, A., Alves, M., Macedo, H. (2016) Vehicle driving analysis in regards to fuel consumption using Fuzzy Logic and OBD-II devices. *EATIS 2016*: 1-4.
- [7] Signoretti, G., Silva, M., Araujo, J., Silva, I., Ferrari, P., Sisinni, E. (2019) A Dependability Evaluation for OBD-II Edge Devices: An Internet of Intelligent Vehicles Perspective. *LADC 2019*: 1-9.